



Embodied AI

Lecture 2: Embodied AI System Overview

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Agenda

- The Embodied AI paradigm
- Embodied AI System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab

What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future
 - General-purpose home robots!



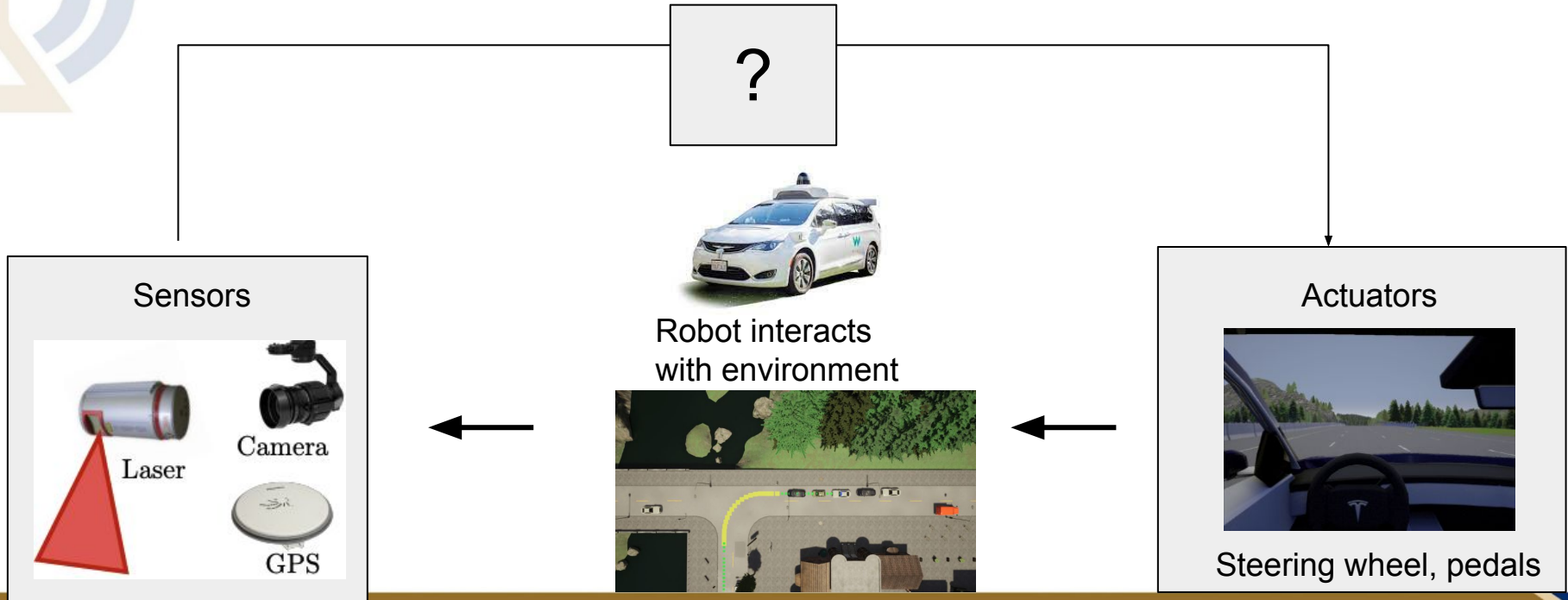
The Embodied AI Paradigm

- Let's start with - How do we make a self-driving vehicle?
 - Given
 - Starting location, Goal location
 - Map of the city
 - Sensors - Camera, LIDAR, etc.
 - Objective
 - Minimize distance to goal
 - Constraint
 - Follow the road
 - Follow traffic rules
 - No crashing into other objects



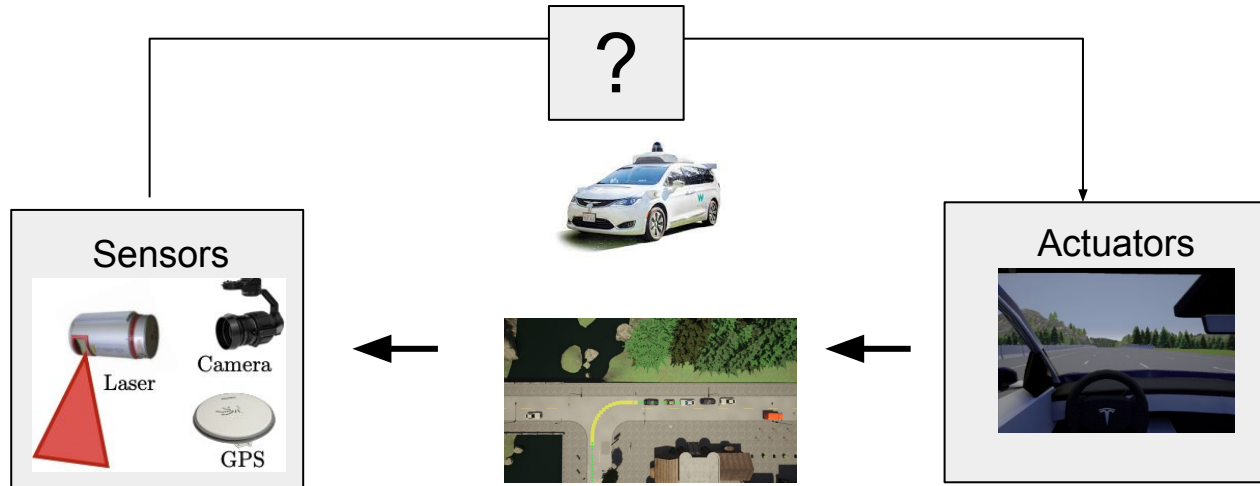
The Embodied AI Paradigm

- How do we make a self-driving vehicle? - here is a flow chart
 - Actions includes steering wheel and pedals



The Embodied AI Paradigm

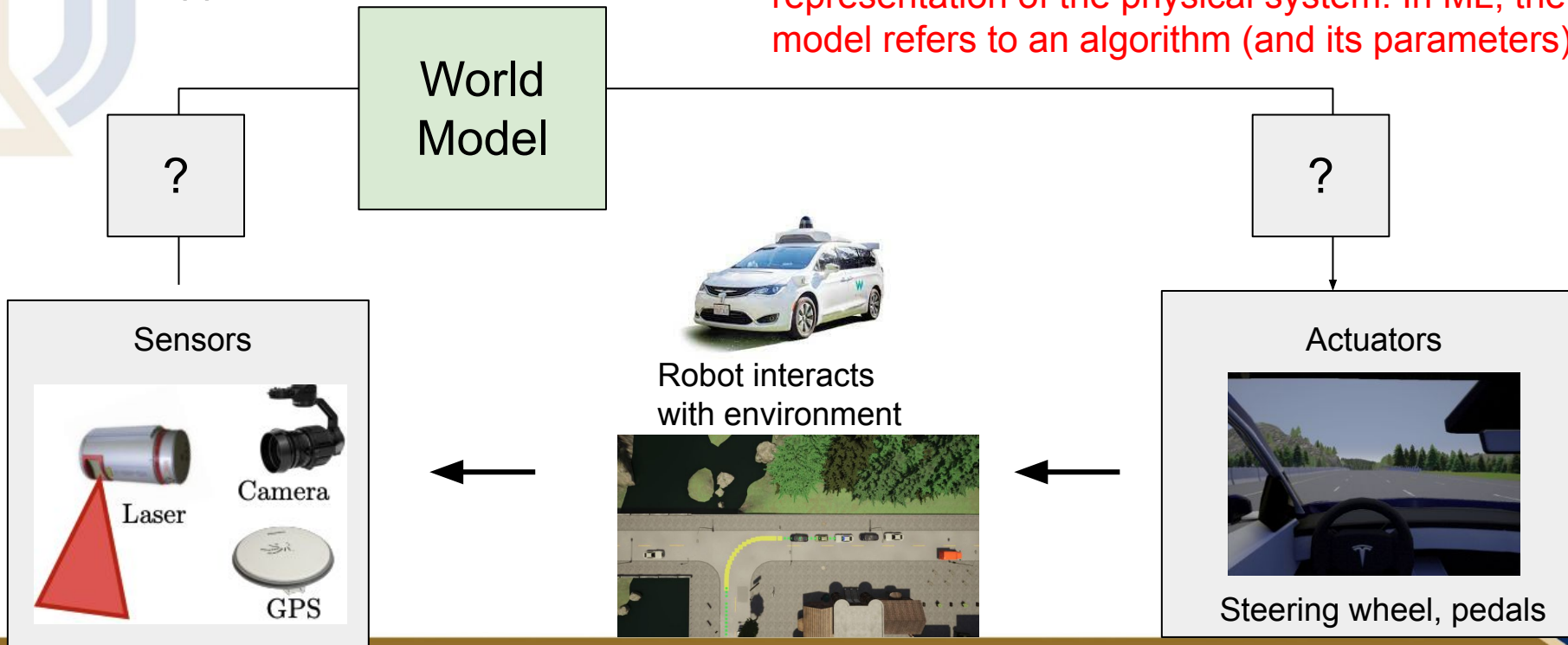
- Key questions
 - Q1: **Assume we know everything about the world.** What commands should we send to the actuator?
 - Q2: How do we update the world knowledge given the sensors?



The Embodied AI Paradigm

- Suppose we have a **world model**

Note in robotics, people use “model” to refer to a representation of the physical system. In ML, the model refers to an algorithm (and its parameters).



The Embodied AI Paradigm

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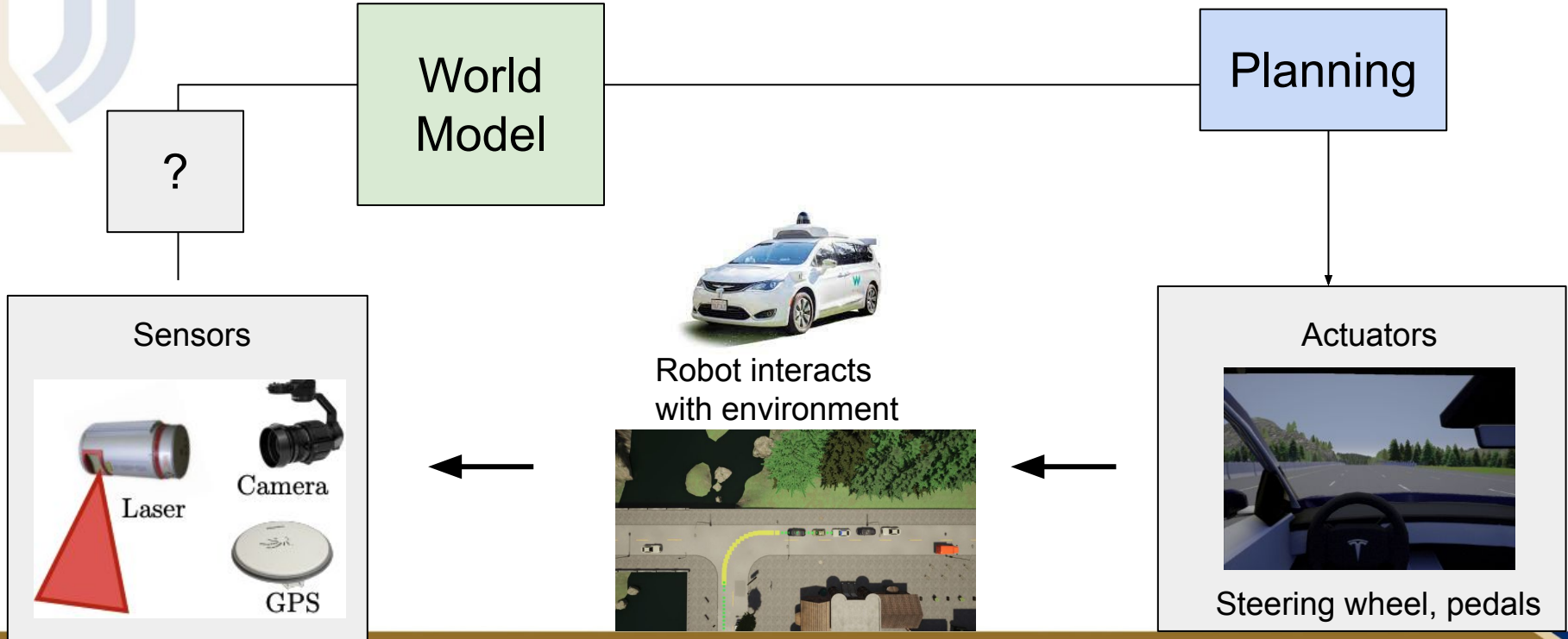
- What is the world model? It includes:
 - **Where is the robot/vehicle? What is its current status?**
 - What are obstacles (sidewalk, grass, trees) and what are available to drive (road)?
 - What objects to avoid (other vehicle, pedestrians)?
 - What are the properties of the vehicle? (Car Model)
 - How fast it can go? Acceleration rate?
 - Wet ground?
 - Nowadays, **world model (or world action model, WAM)** means “pixel and action prediction” model. So the vision generative models are considered world model

To be more precise, world model means a “state transition” model $P(s_t, a_t)=s_{t+1}$
It predicts the future “state”.

“WAM” refers to world action model $P(s_t, a_t)=(s_{t+1}, a_t)$, it predicts both future states and actions

The Embodied AI Paradigm

- Given the world knowledge, we can then do planning!



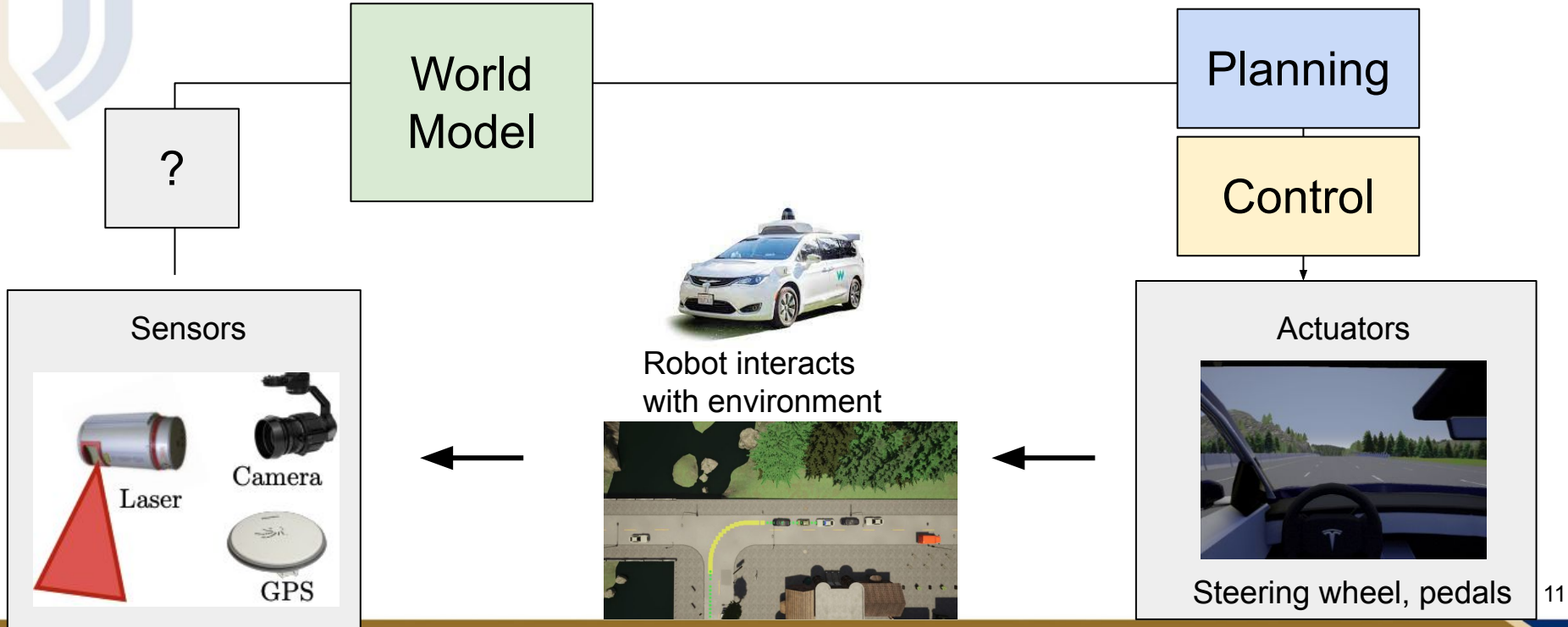
The Embodied AI Paradigm

- What is planning?
 - **Planning is an optimization problem in which we find the optimal sequence of actions towards minimizing a cost function**
 - In the self-driving case, minimizing the robot's distance to the goal location
 - While satisfying constraints (no crashing, follow traffic rules)
 - There are global planning (global route) and local planning (change lane? Turn?)

The Embodied AI Paradigm

- Given the plan, what about the control?

The plan gives where to go. Now we need to control the robot to do so



The Embodied AI Paradigm

- What is planning and **control**?
 - There are global (global route) and local planning (change lane? Turn?)
 - **A controller will execute the plan by giving specific commands to the actuators**
 - For example, convert the trajectory/motion plan (which way the car should go and how fast) into the **steering** and **pedal** command

The Embodied AI Paradigm

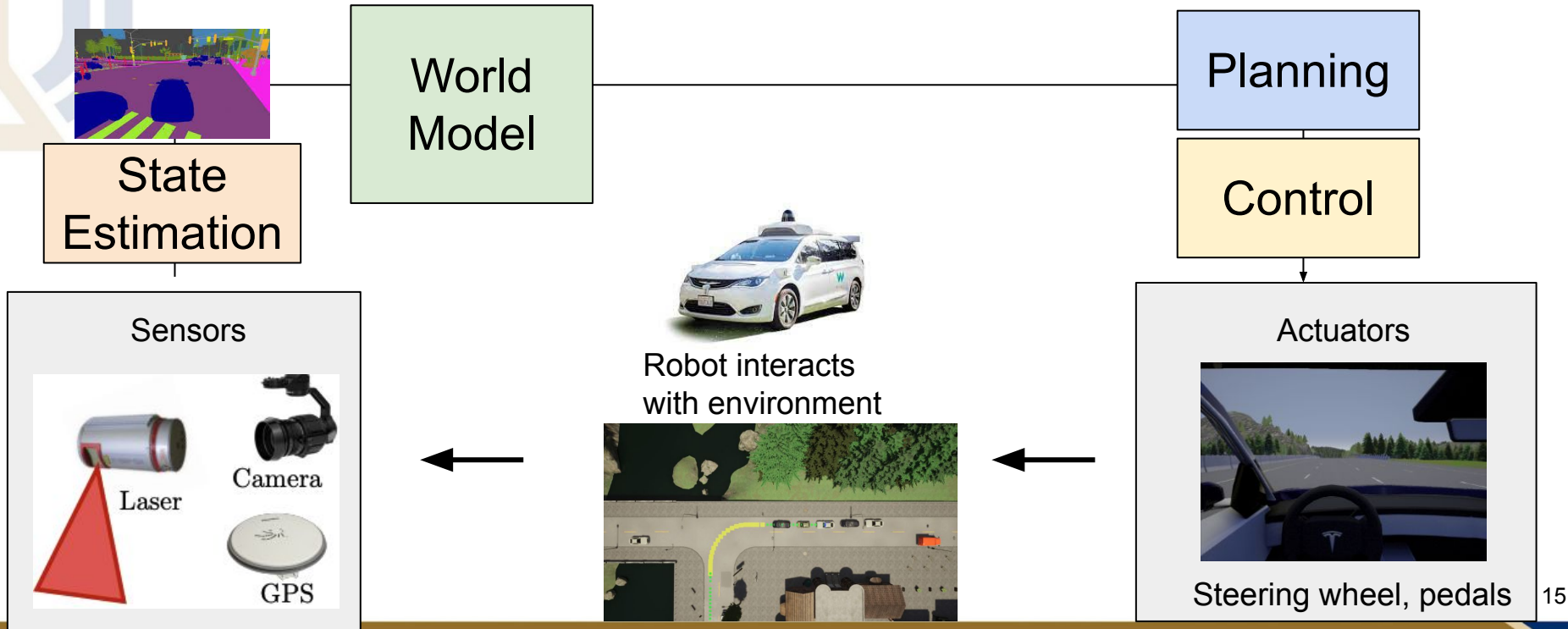
- Key questions
 - Q1: Assume we know everything about the world. What commands should we send to the actuator?
 - Done. We know how to control the car if we know about the world
 - **Q2: How do we update the world knowledge given the sensors?**

The Embodied AI Paradigm

- **State Estimation** - Update the world knowledge given the sensors
 - Given the sensor data, the perception module
 - extracts necessary information (about the surroundings)
 - localizes the robot in the world

The Embodied AI Paradigm

- How do we make a self-driving vehicle? - Here is the (simplified) completed graph



CMU's winning entry to DARPA Self-Driving challenge 2007

- Let's look at a successful self-driving system from researchers at CMU





Tartan Racing: A Multi-Modal Approach to the DARPA Urban Challenge

April 13, 2007

Chris Urmson, Joshua Anhalt, Drew Bagnell, Christopher Baker, Robert Bittner,
John Dolan, Dave Duggins, Dave Ferguson, Tugrul Galatali, Chris Geyer, Michele Gittleman,
Sam Harbaugh, Martial Hebert, Tom Howard, Alonzo Kelly, David Kohanbash, Maxim Likhachev,
Nick Miller, Kevin Peterson, Raj Rajkumar, Paul Rybski, Bryan Salesky, Sebastian Scherer,
Young Woo-Seo, Reid Simmons, Sanjiv Singh, Jarrod Snider, Anthony Stentz,
William “Red” Whittaker, and Jason Ziglar

Carnegie Mellon University

Hong Bae, Bakhtiar Litkouhi, Jim Nickolaou, Varsha Sadekar, and Shuqing Zeng

General Motors

Joshua Struble and Michael Taylor

Caterpillar

Michael Darms

Continental AG

CMU's winning entry to DARPA Self-Driving challenge 2007

- See the number of sensors on top of the vehicle
- And the 3D visualization of mapping and localization of the car

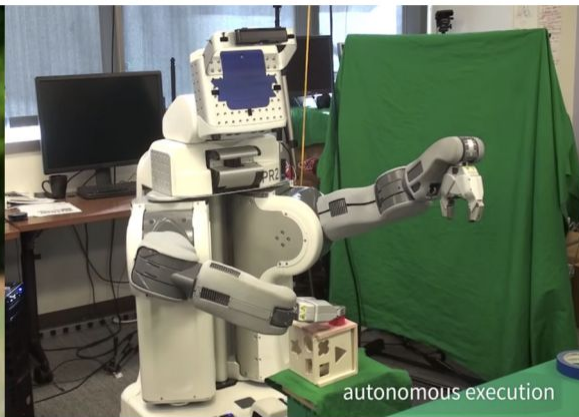


The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - Drones, robotic arms, legged machines



[Sa et al. IROS 2014]



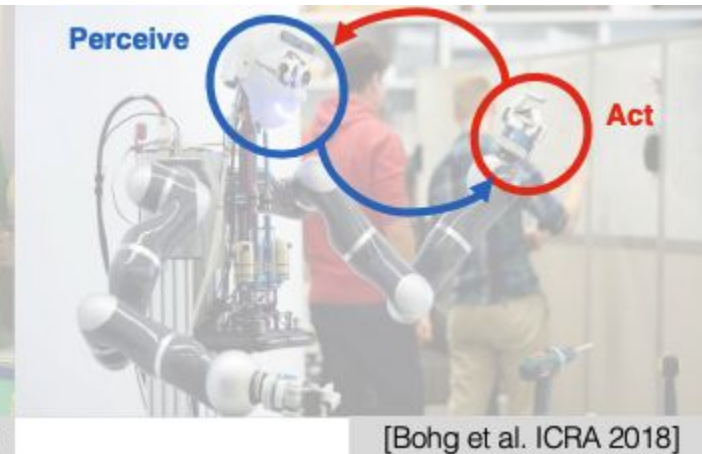
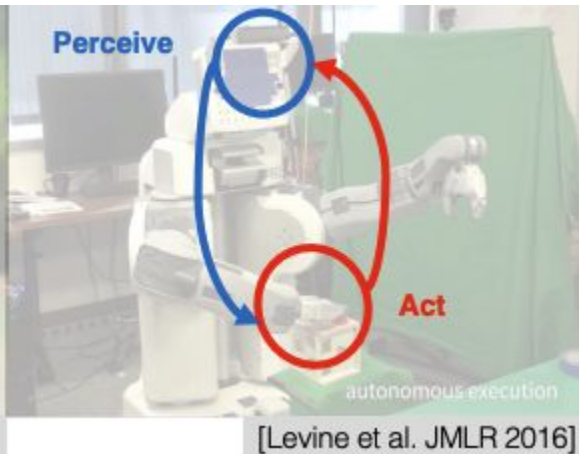
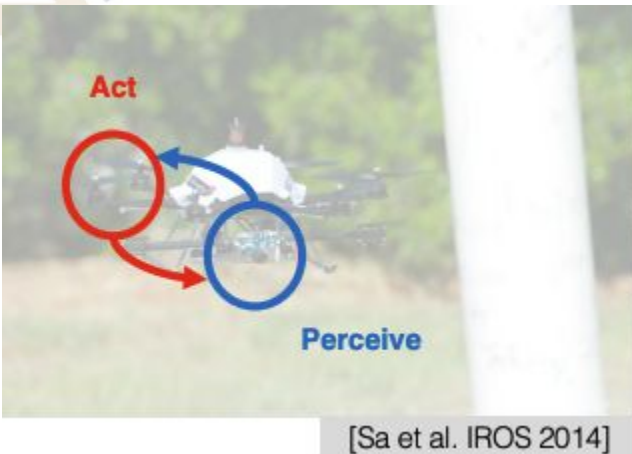
[Levine et al. JMLR 2016]



[Bohg et al. ICRA 2018]

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - The key is the perception-action loop

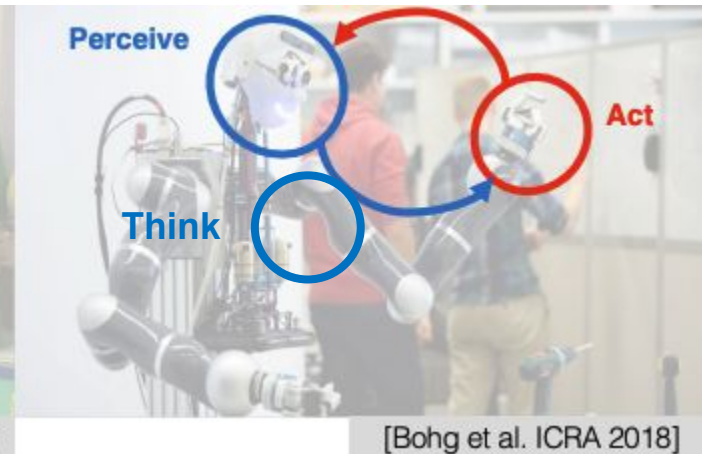
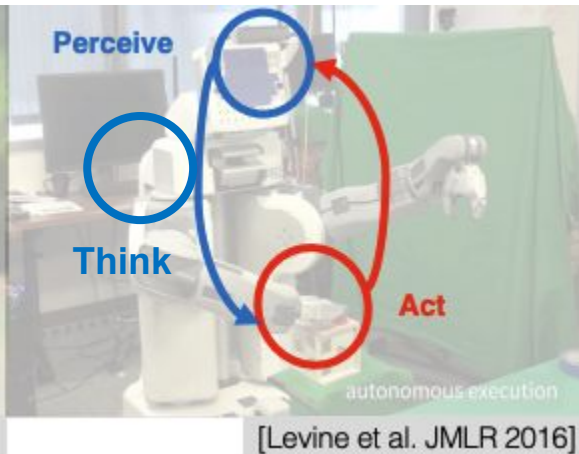
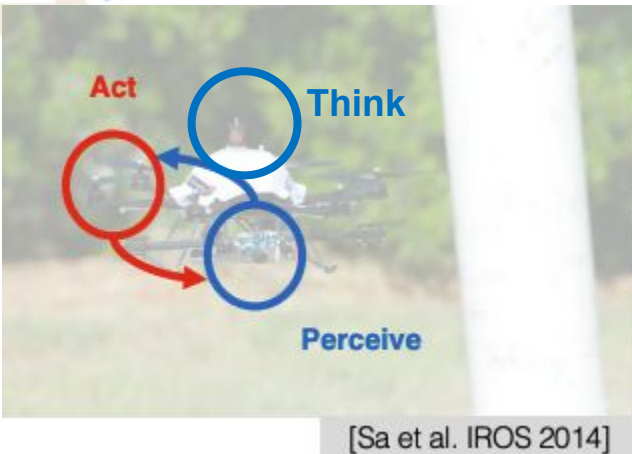


Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)

Act - Actuators: Joints, arms, claws, motors

The Embodied AI Paradigm

- Besides self-driving vehicle, we can also generalize this paradigm to other robot systems
 - Adding **machine learning** to close the perception-action loop (we call the process “Think”)

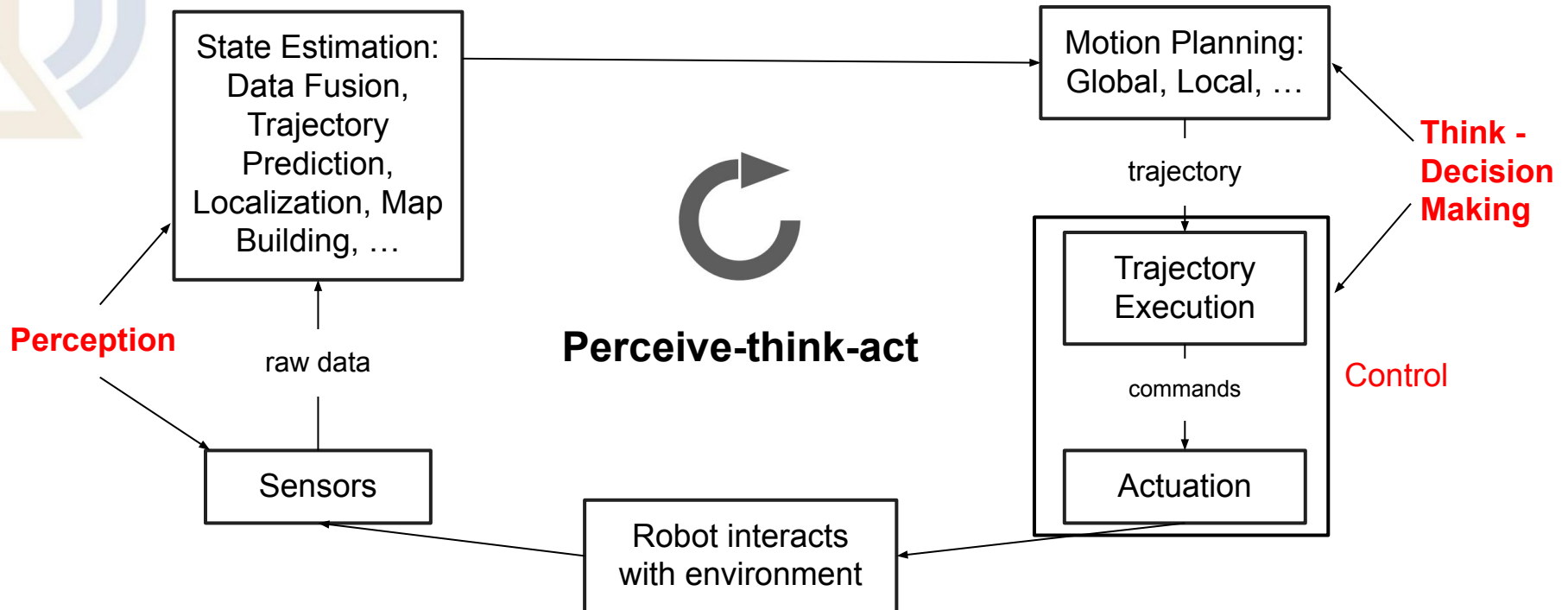


Perceive - Sensors are similar: Cameras, LIDARs, Inertial Measurement Units (IMUs)

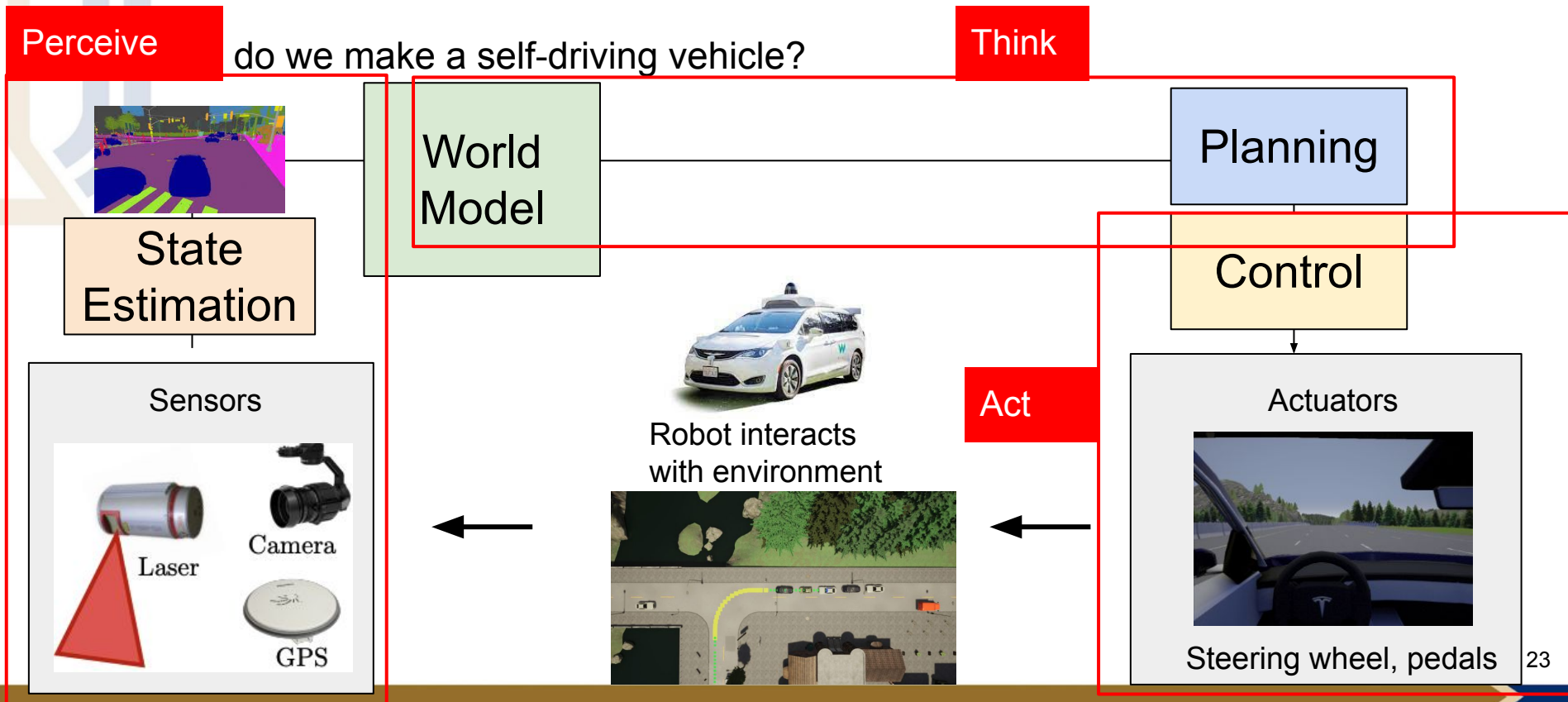
Act - Actuators: Joints, arms, claws, motors

The Embodied AI Paradigm

- A more generalized paradigm - The Perceive-think-act circle



The Perceive-Think-Act Cycle in previous self-driving system

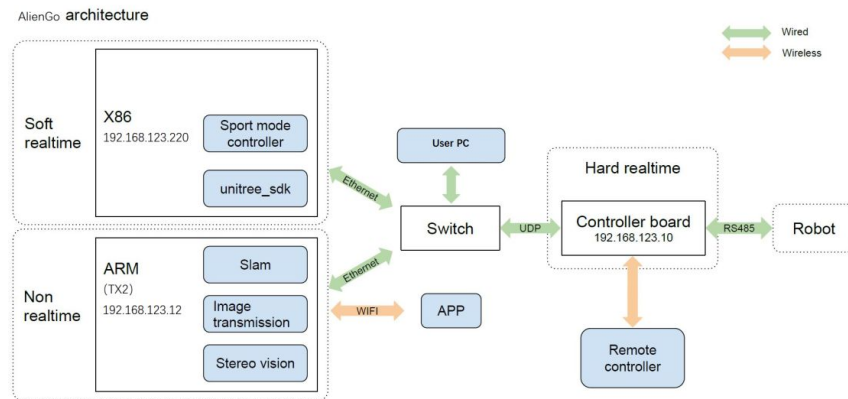


Agenda

- The Embodied AI paradigm
- **Embodied System Design**
- Sensors Overview
- Robot Vision vs. Computer Vision

Embodied System Design

- The hardware
 - Sensors
 - Actuators
- The software
 - Lowlevel: Control Unit Drivers
 - Middleware: Robot Operating System (ROS)
 - Based on Data Distribution Service (DDS)
 - Communication between modules
 - Data logging
 - Visualization
 - Perception models
 - Motion planning models
- Principles
 - Follow the Perceive-think-act paradigm
 - Modular, robust, data logging and playback

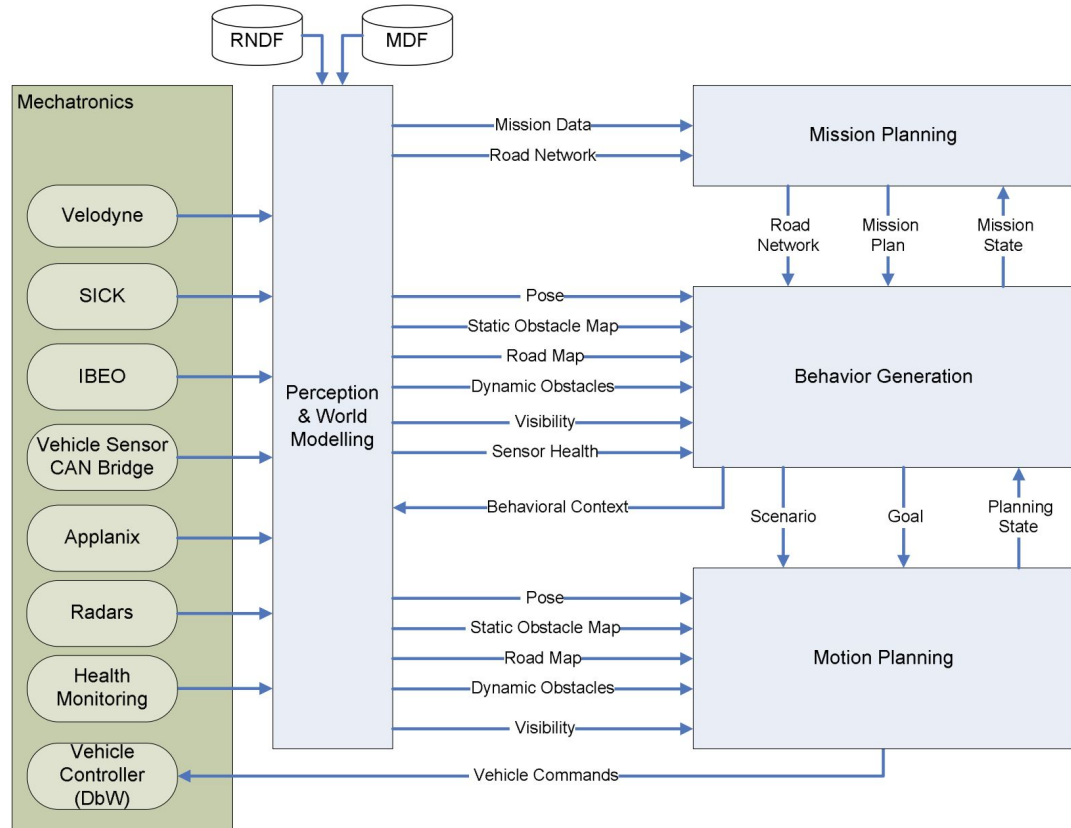


The architecture of Unitree's four legged-robot

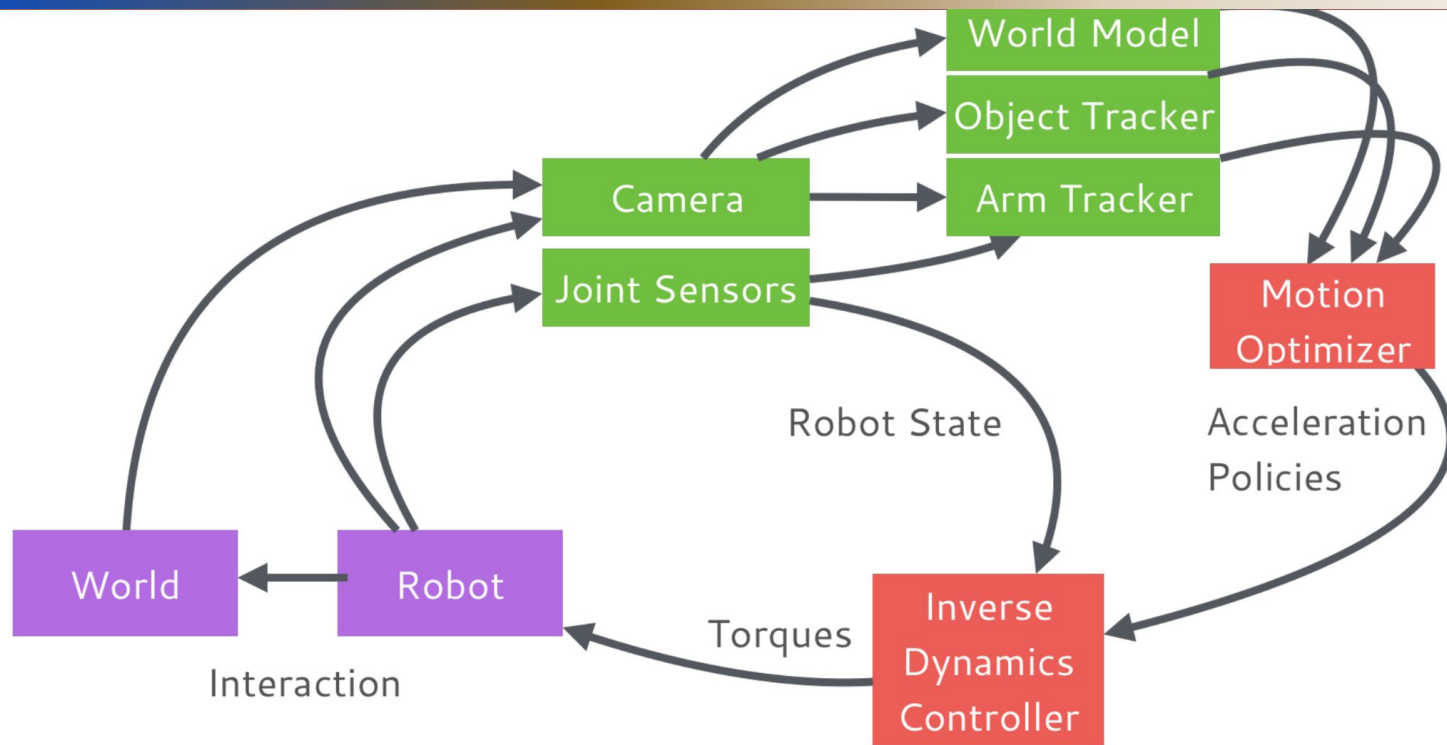
Embodied System Design

- Example: CMU's Self-driving system

- Perception
- Planning
- Mechatronics (Controller & sensors)



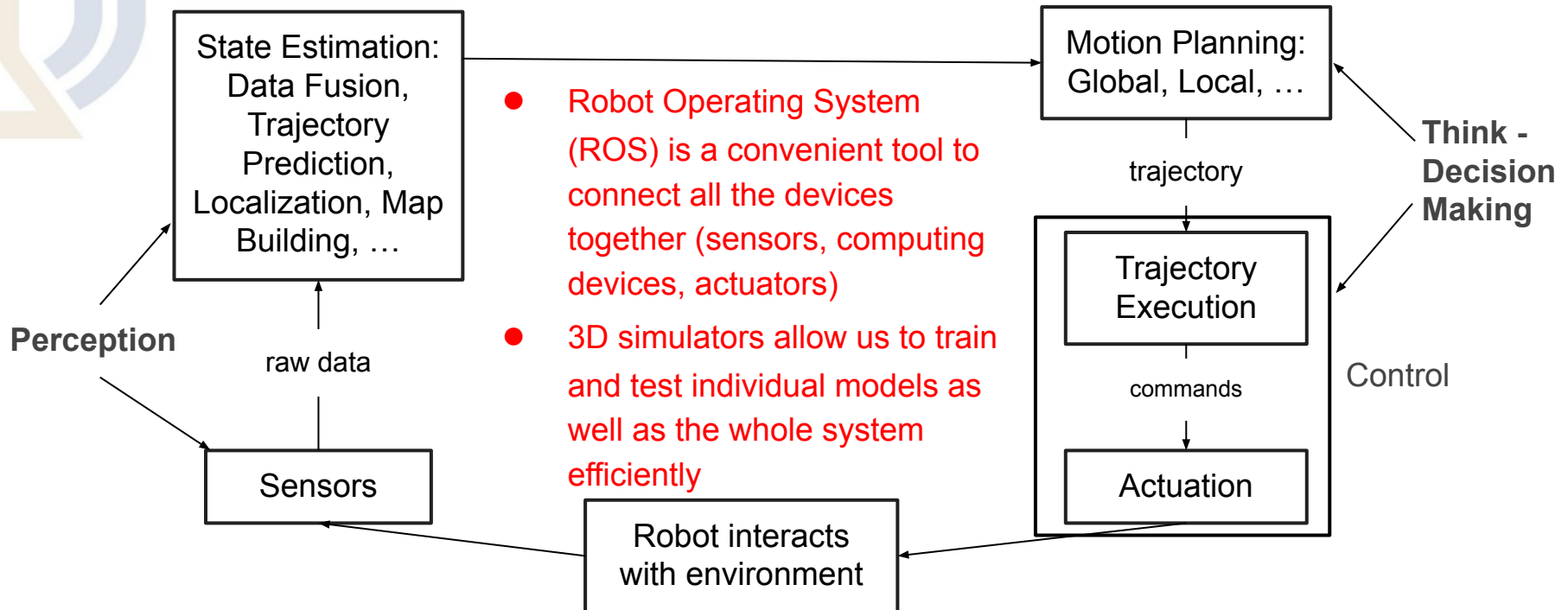
Embodied System Design



- Example: Kappler et. al. Real-Time Perception Meets Reactive Motion Generation. ICRA 2018

Embodied System and The Embodied AI Paradigm

- ROS and 3D Simulator are key component for Embodied AI

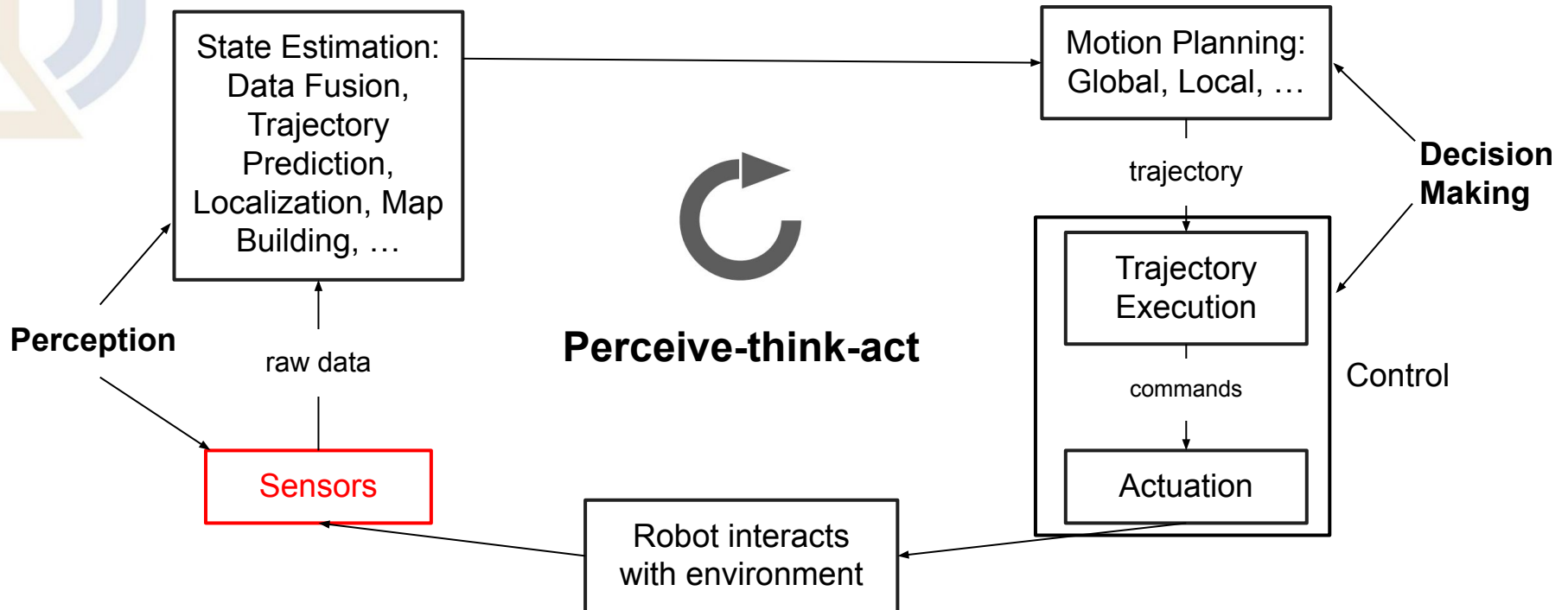


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- The Embodied AI paradigm
- Embodied System Design
- **Sensors Overview**
- Robot Vision vs. Computer Vision

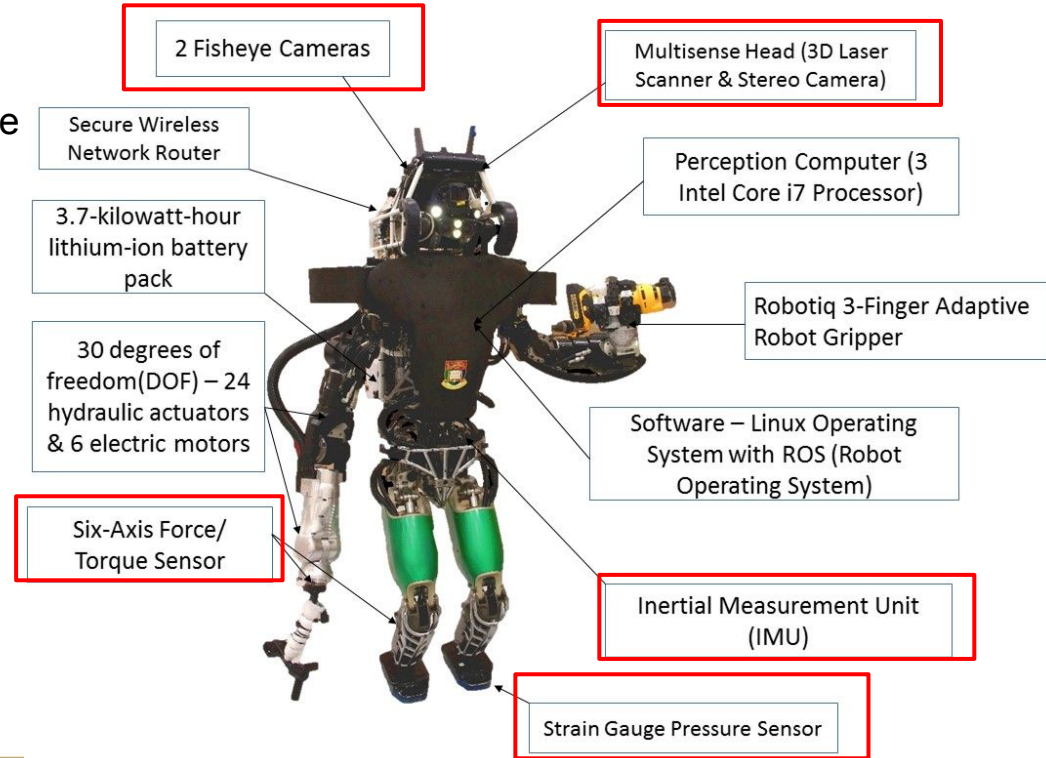
The Embodied AI Paradigm

- The Perceive-think-act circle



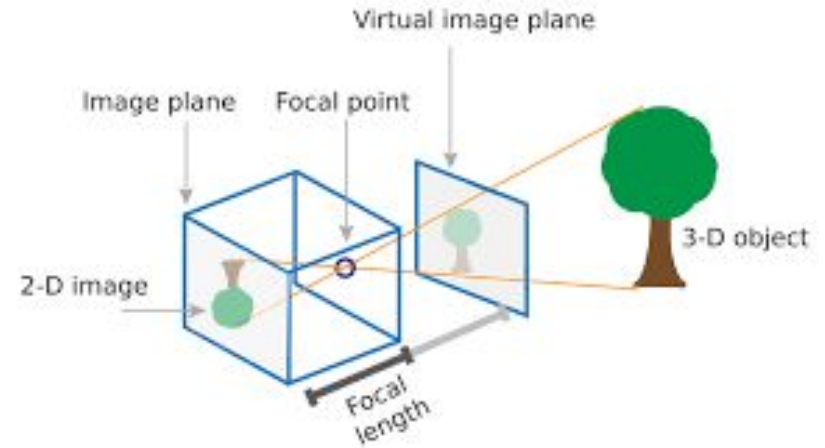
Robotic Sensors

- Multimodal sensors are key to robot perception
 - Example: DARPA robotic challenge 2015
 - Fisheye cameras
 - LIDAR
 - Inertial Measurement Unit (IMU)
 - Torque Sensor
 - Pressure Sensor



Robotic Sensors

- Multimodal sensors are key to robot perception
 - Cameras, Stereo Cameras
 - How does the 3D world project to a 2D image?
 - We will learn about it in lecture 4!

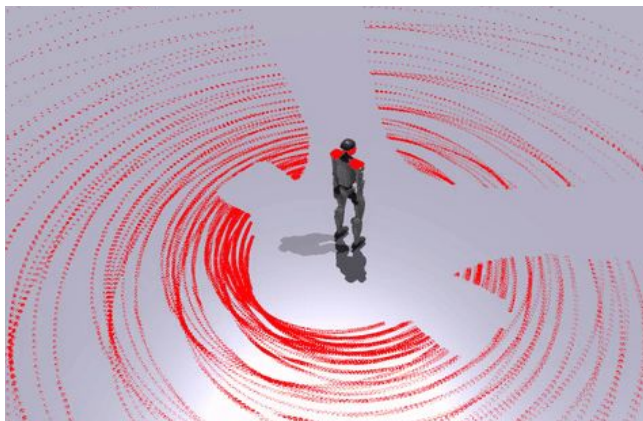


Robotic Sensors

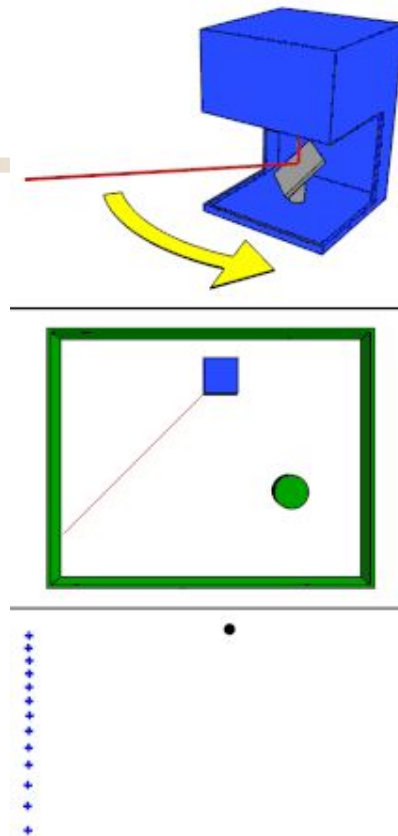
- Multimodal sensors are key to robot perception
 - Lidar
 - Short for light detection and ranging
 - Send laser and get distance to obstacles
 - Why the discontinuance?
 - You can only sample finite angles



LiDAR Sim from CARLA

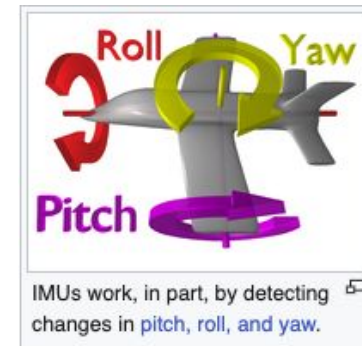
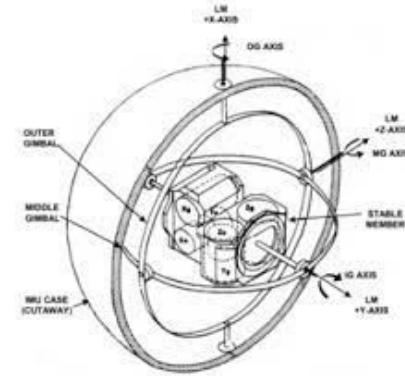
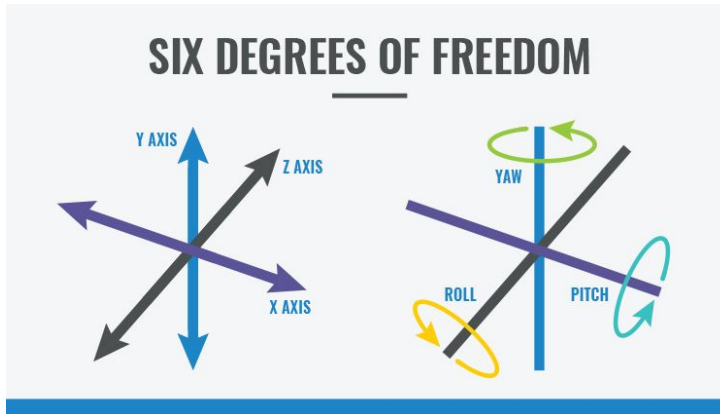


LiDAR Sim on robot from our work: OmniPerception-CoRL'25



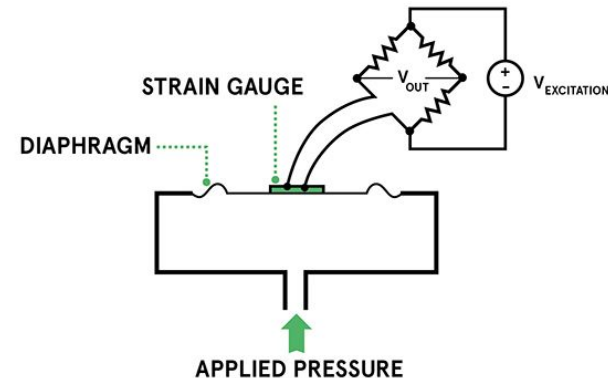
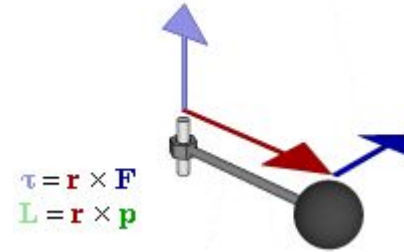
Robotic Sensors

- Multimodal sensors are key to robot perception
 - Inertial Measurement Unit (IMU)
 - Measure angular rate, orientation, acceleration, etc.
 - Used in drones, and smartphones!



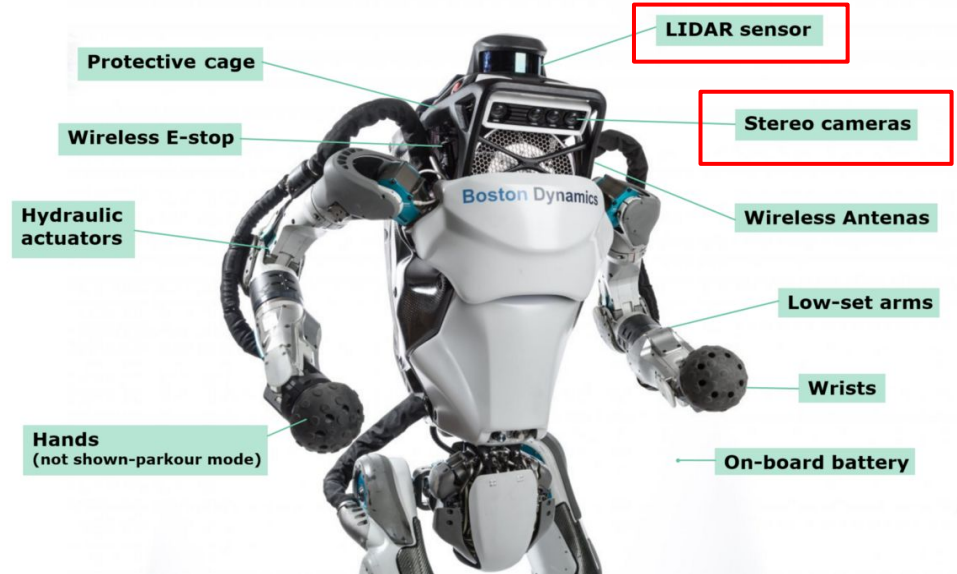
Robotic Sensors

- Multimodal sensors are key to robot perception
 - Torque Sensor
 - Measure rotation force
 - Strain Gauge Pressure Sensor
 - Apply pressure and electronic signal changes
 - Touch sensor!



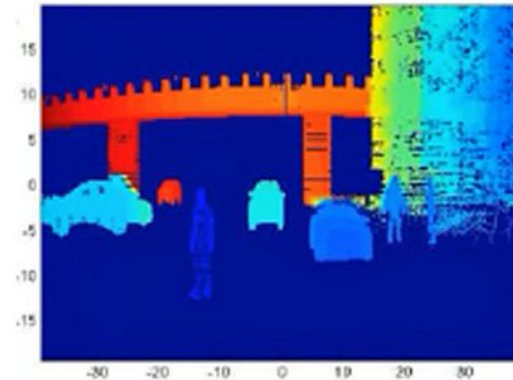
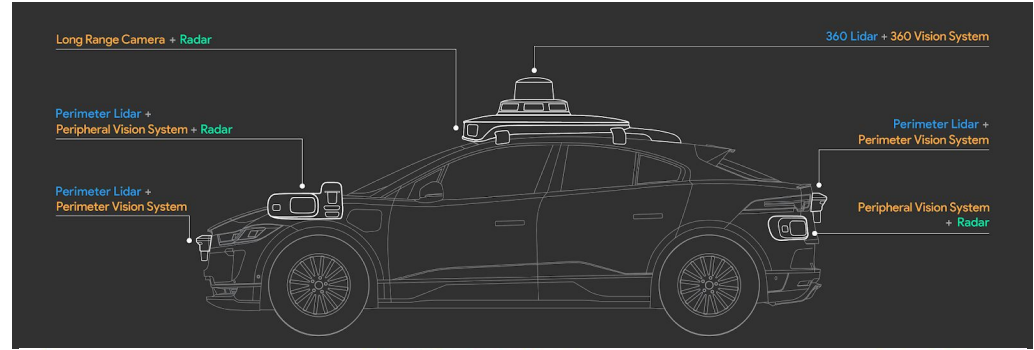
Robotic Sensors

- Multimodal sensors are key to robot perception
 - Boston Dynamics Atlas
 - Cameras
 - Lidar
 - Inertial Measurement Unit (IMU)?

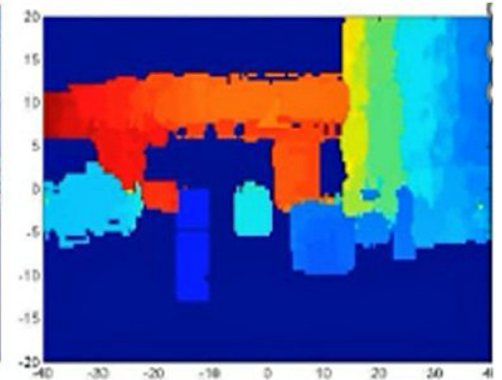


Robotic Sensors

- Multimodal sensors are key to robot perception
 - Waymo Self-driving Cars
 - Cameras
 - Lidar
 - Uses near-infrared
 - More precise
 - Radar
 - Uses microwave
 - Longer range



Lidar



High Resolution Radar

Robotic Sensors

- Multimodal sensors are key to robot perception
 - In 3D simulator CARLA
 - RGB Camera, IMU, Depth Camera, GPS, Radar, Lidar (you can set Lidar properties), Event Camera, etc.
 - Event camera: basically unlimited FPS (low-resolution) camera



Agenda

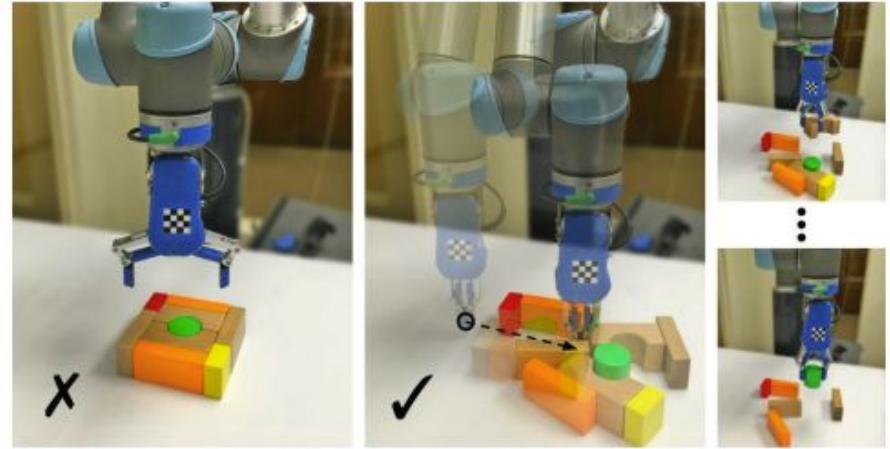
- The Embodied AI paradigm
- Embodied System Design
- Sensors Overview
- Robot Vision vs. Computer Vision

Robot Vision vs. Computer Vision

- Robot vision is embodied, active and environmentally situated



[Detectron - Facebook AI Research]



[Zeng et al., IROS 2018]

Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
- **Embodied**: Robots have physical bodies and experience the world directly. Their actions are part of a dynamical system together with the environment and have **immediate feedback** on their own sensation
- **Active**: Robots are **active perceivers**. They should know what and why it wishes to sense, and they should be able to choose what to perceive, and determine how, when and where to achieve that perception
- **Situated**: Robots are situated in the world. The environment will affect the whole system

Robot Vision vs. Computer Vision

- Robot vision is **embodied, active** and **environmentally situated**
 - Like human ego-vision



Agenda

- The Embodied AI paradigm
- Embodied System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Recent robot demos from our lab

具身智能实验室 / Precognition Lab Overview

小红书

扫描二维码
在小红书找到我

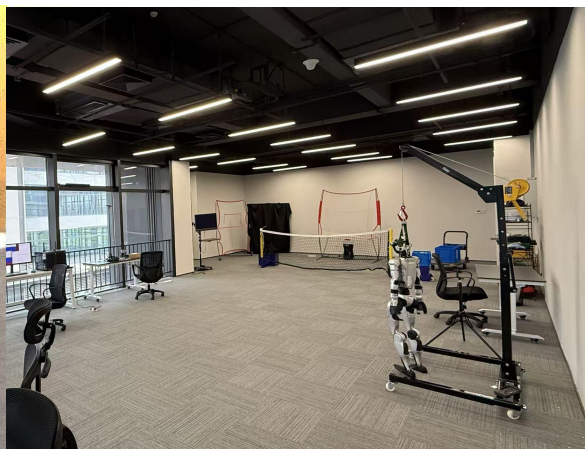
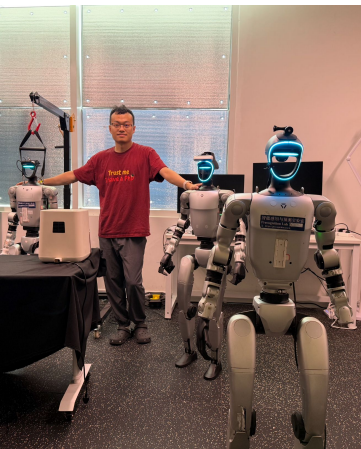


People & Funding Projects & Lab

- **PI: Junwei Liang, Asst. Prof., CMU PhD.**
 - CoRL/NeurIPS Area Chair, CoRL GenPriors Best Paper
- Projects: NSFC, Tencent Robotics X, Meituan Robotics, Alibaba, Huawei, etc.
 - **Totaling 5M+ RMB**
- Lab: **200+ Sq Meters lab** for robot experiments
- 10 PhD students. **Soon-graduates got 3-9M RMB offer**

Hardware

- We have a total of 80 GPUs
 - A6000/**Pro 6000** GPU servers
 - L40/4090 workstations
- **Robot Platforms**
 - Humanoid robots (n* Unitree G1 and H2)
 - Four-legged robots (A2 and Go2-W)



具身智能实验室 / Precognition Lab - Overview

- Goal: Build human-level AI systems that could effectively perceive, reason and interact with the real-world for the good of humans
 - So eventually we have a **general household robot** that can help us with everyday tasks

Spatial Intelligence → VLM

Mobility - Navigation + Locomotion

“Action”

Manipulation

“How to do stuff”

“How to get somewhere”



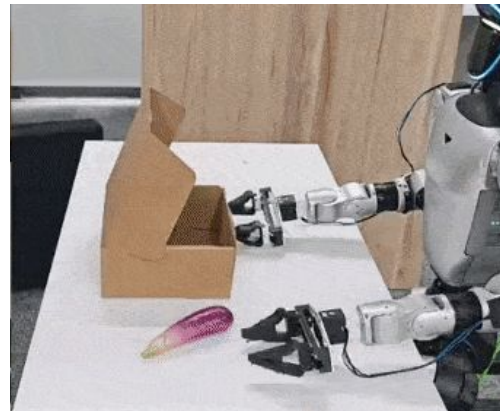
Wheeled



Quadrupedal

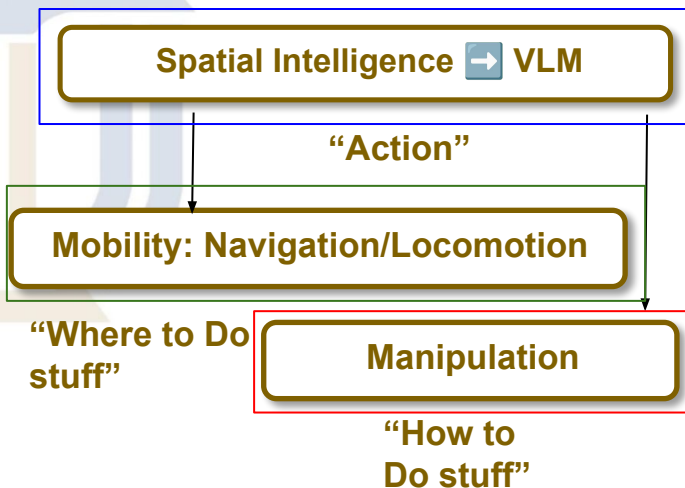


Humanoid



Lab website: <https://precognition.team/>

Research Highlights of Precognition Lab



*We show these demo today

Towards General Service Embodied AI

- 3D Visual Grounding - SeeGround (CVPR'25, NeurIPS'25)
- **Social Navigation - FLUX (*ICRA'25, IROS'26, RA-L'26)**
- LiDAR-based WBC Locomotion (CoRL'25 Oral)
- **Perceptive Behavior Foundation Model (CoRL'26)**
- Imitation learning for Manipulation - GLOVER (CoRL'25)
- VLA Manipulation Benchmark - AGNOSTOS (NeurIPS'25)
- **Motion World Action Model (*'26)**
- **End-to-end SpeechVLA Humanoid Agent (*'26 Deployed at Uni. Visitor Center)**



香港科技大學(廣州)

THE HONG KONG UNIVERSITY OF SCIENCE
AND TECHNOLOGY (GUANGZHOU)



From Cognition to Precognition: A Future-Aware Framework for Social Navigation 基于未来预测的社交导航系统

Zeying Gong¹, Tianshuai Hu², Ronghe Qiu¹, Junwei Liang^{1,2*}

¹ AI Thrust, The Hong Kong University of Science and Technology (Guangzhou)

² Department of Computer Science and Engineering, The Hong Kong University of
Science and Technology

*Corresponding author

**All authors are my students and
I'm the only advisor.**

Work done in almost 8 months.

<https://zeying-gong.github.io/projects/falcon/>

Social Navigation - 社交导航主题研究 - ICRA 2025

Third-party Media report: 量子位报道: https://mp.weixin.qq.com/s/NT8Wk6N3S_mQwN_f39Arog

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction [3]

- Predict human trajectory using visual information^[1,2], make sure robot plans a path that follows social etiquette. Ensure physical and mental safety of AI robots
- **The robot is going to the plant.** See here how the robot **moves backward** to avoid the person



[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.

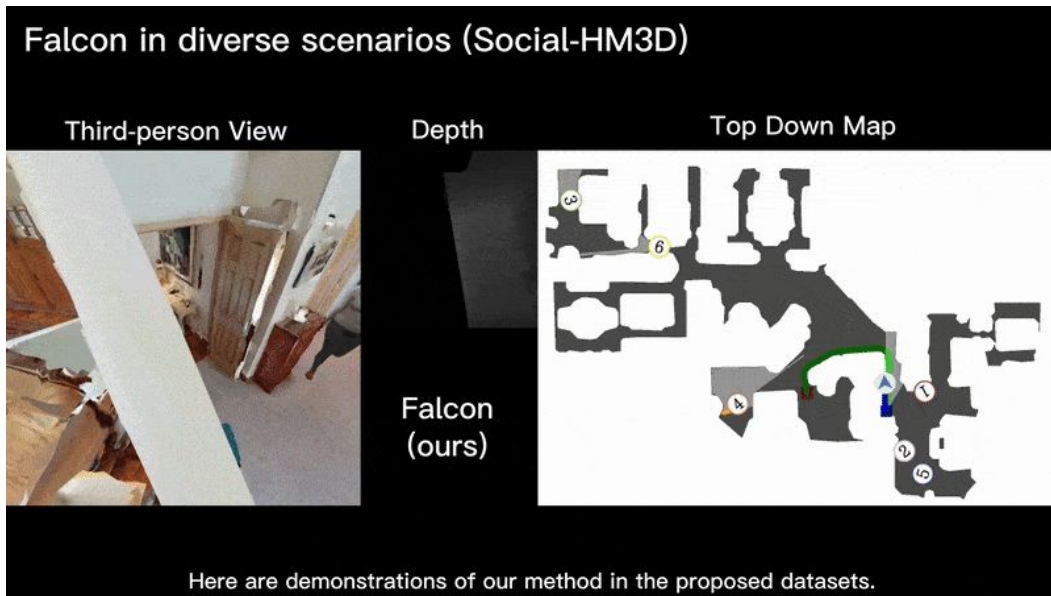
[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019

[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

Our robot demos: Social Navigation

Social Navigation using Human Trajectory Prediction

- More examples - You will be able to train this model yourselves for Project 3 homework!



[1] Liang, Junwei, et al. "The garden of forking paths: Towards multi-future trajectory prediction." CVPR 2020.

[2] Liang, Junwei, et al. "Peeking into the future: Predicting future person activities and locations in videos." CVPR 2019

[3] "From Cognition to Precognition: A Future-Aware Framework for Social Navigation", ICRA 2025

Real-World Experiment on a Humanoid Robot (Unitree G1):

FLUX on a Humanoid Robot (4x Speed)



Predicted Traj.

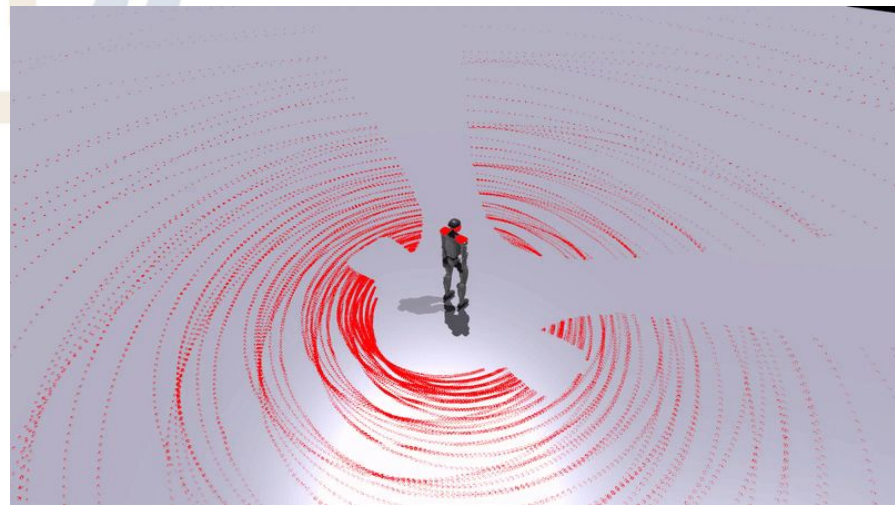
Selected Traj.

RGB

Depth

Our robot demos: Agile Locomotion

- We can also utilize reinforcement learning with 3D simulator to train a “locomotion policy” that allows robot to avoid moving obstacles



*This is 0.5x playback speed

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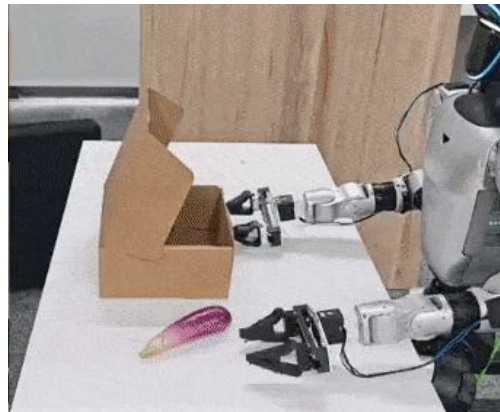
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Quadrupedal



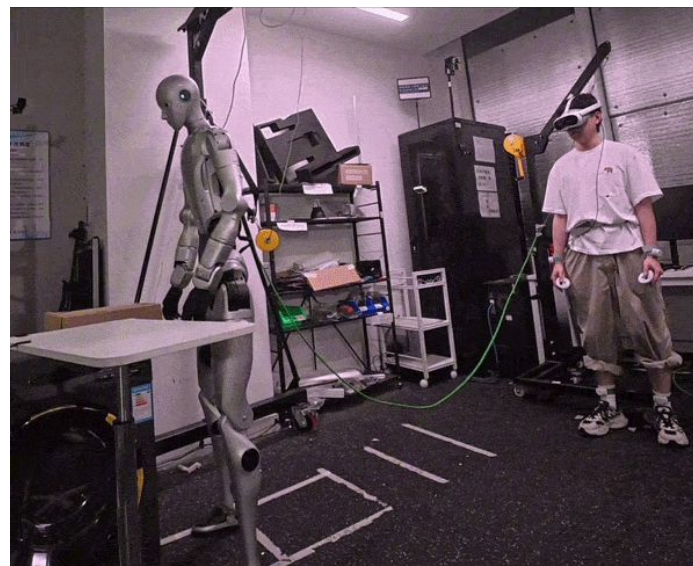
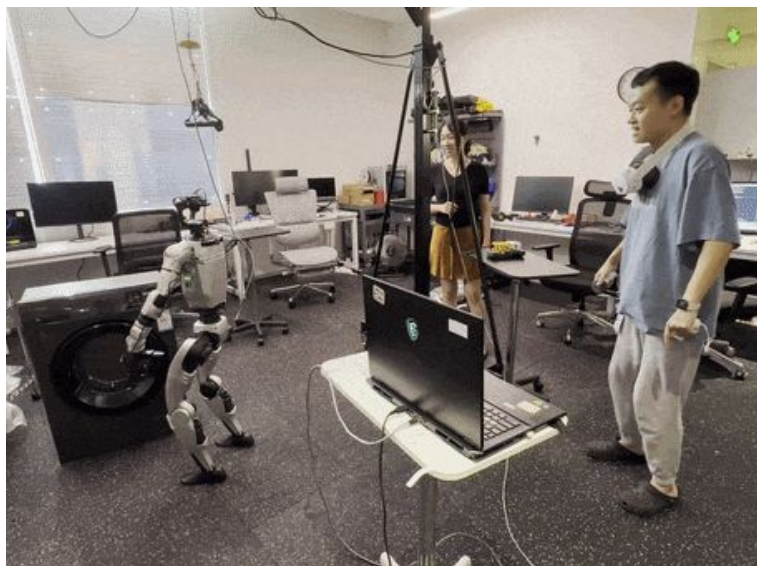
Humanoid



Lab website: <https://precognition.team/>

Whole-body-control TeleOperation

- Teleoperation-Based Data Collection: Acquiring whole-body control data requires human operators to map joint-level commands via VR, which inherently necessitates a robust baseline foundation model
- The collected robot data with vision data can be used to train VLA models





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MotionWAM: Towards Foundation World Action Models for Real-Time Humanoid Loco-Manipulation 基于**视频生成世界模型**的人形机器人**整身操作**

Jia Zheng^{1,2,†} Teli Ma^{1,2,†} Yudong Fan¹ Zifan Wang^{1,2} Shuo Yang^{1,*} Junwei Liang^{2,3,*}

¹Mondo Robotics ²HKUST(GZ) ³HKUST

[†]Equal contribution. ^{*}Corresponding author, Co-advising.

<https://dit4dit.github.io/MotionWAM/>

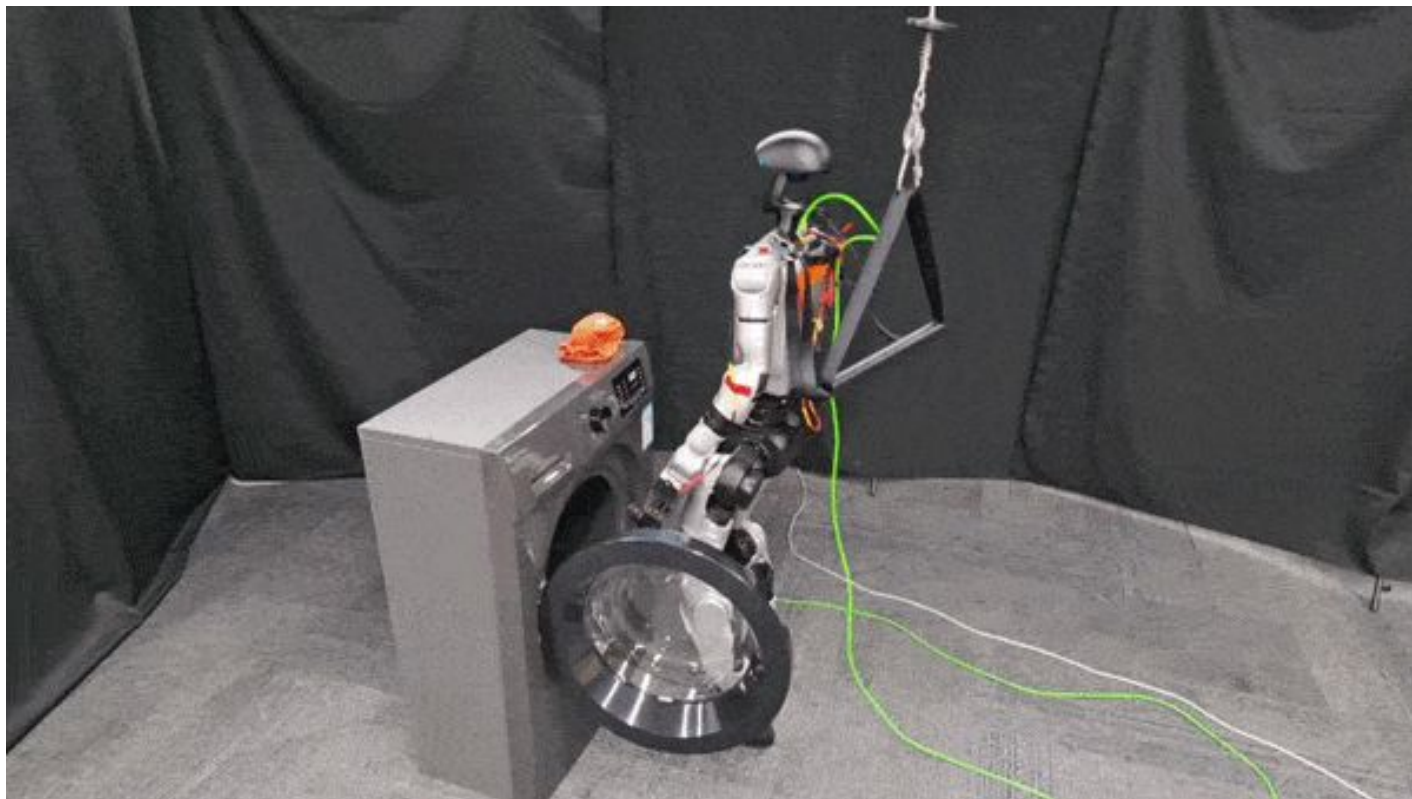
Robotic Manipulation - **机器人操作** 研究 - ArXiv 2026
多家公众号争先报道, 包括具身智能研究室、具身智能之心等

Real robot experiments - Do laundry

- The policy even realizes that the door has not been properly closed and fixes it

Fully autonomous,
no speed-up, no
teleoperation.

Our
MotionWAM-2.5B
can run at 5Hz on a
A100 GPU to
decode a 100-step
action chunk

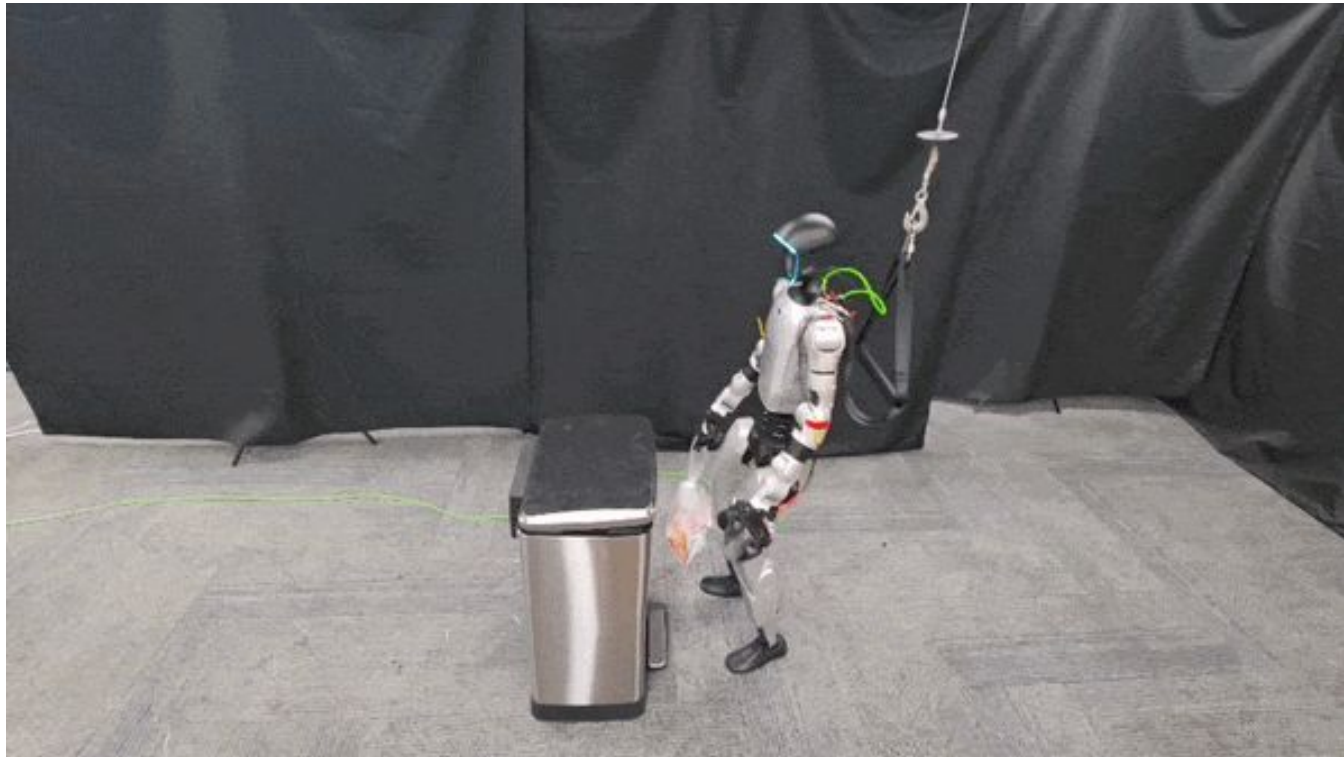


Real robot experiments - Throw away trash

- The robot has to learn precise foot placement

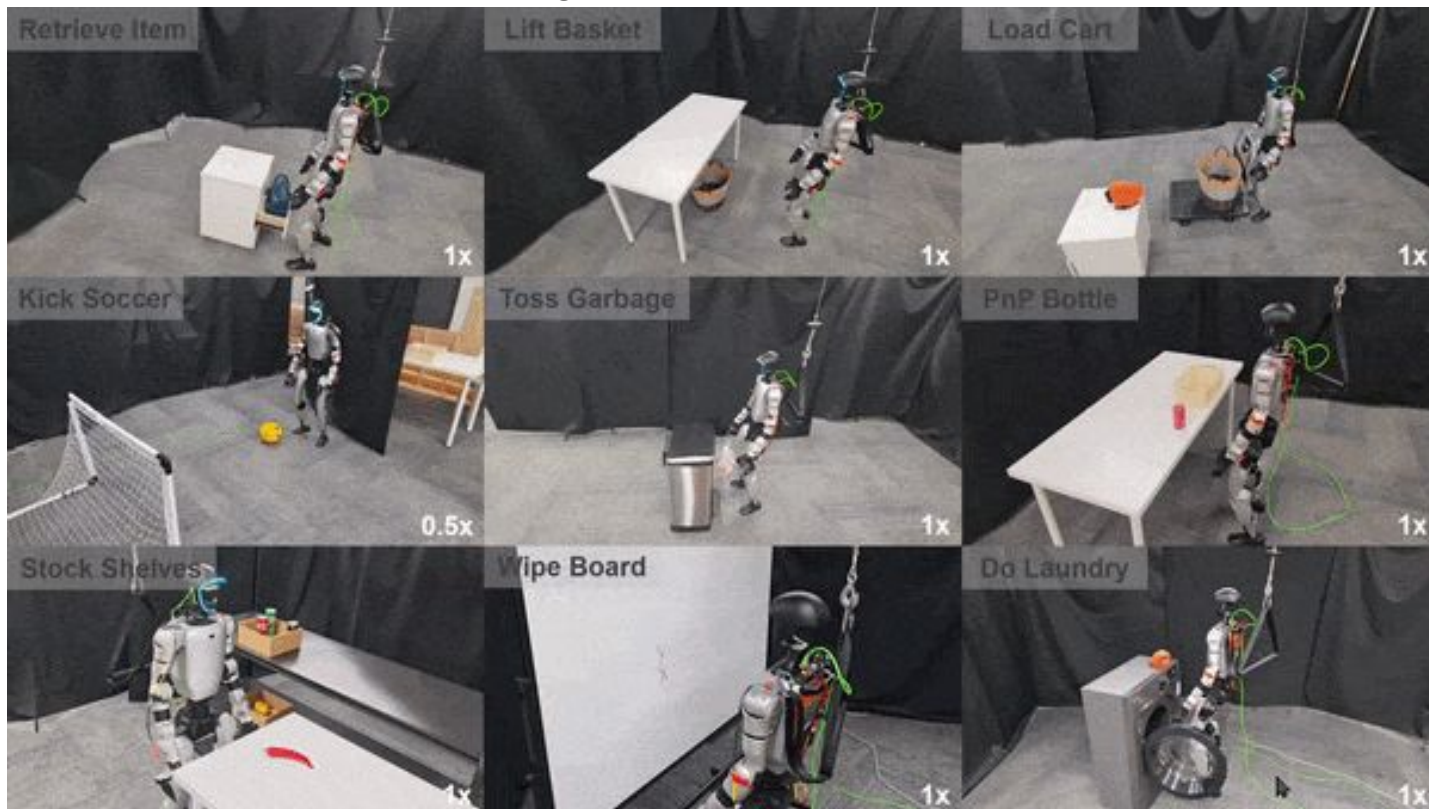
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All 9 Real World Tasks

- More videos at: <https://dit4dit.github.io/MotionWAM/>



具身智能实验室 / Precognition Lab - Overview

- Goal: Build human-level AI systems that could effectively perceive, reason and interact with the real-world for the good of humans
 - So eventually we have a **general household robot** that can help us with everyday tasks

Spatial Intelligence → VLM

Mobility - Navigation + Locomotion

“Action”

Manipulation

“How to get somewhere”



Wheeled



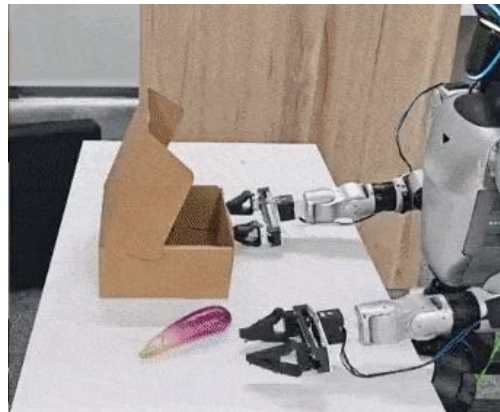
Quadrupedal



Humanoid



“How to do stuff”



Lab website: <https://precognition.team/>

End-to-end SpeechVLA Humanoid Agent (Deployed at Uni. Visitor Center)

- Our demo video for Gaokao has received 150k+ views on bilibili
 - <https://www.bilibili.com/video/BV1U67V6PEGc/>



*check out the full
demo without cuts:
[here](https://www.bilibili.com/video/BV1U67V6PEGc/)

Agenda

- The Embodied AI paradigm
- Embodied AI System Design
- Sensors Overview
- Robot Vision vs. Computer Vision
- Robot Demos from my lab
- Next week - Machine Perception!

Reading Material

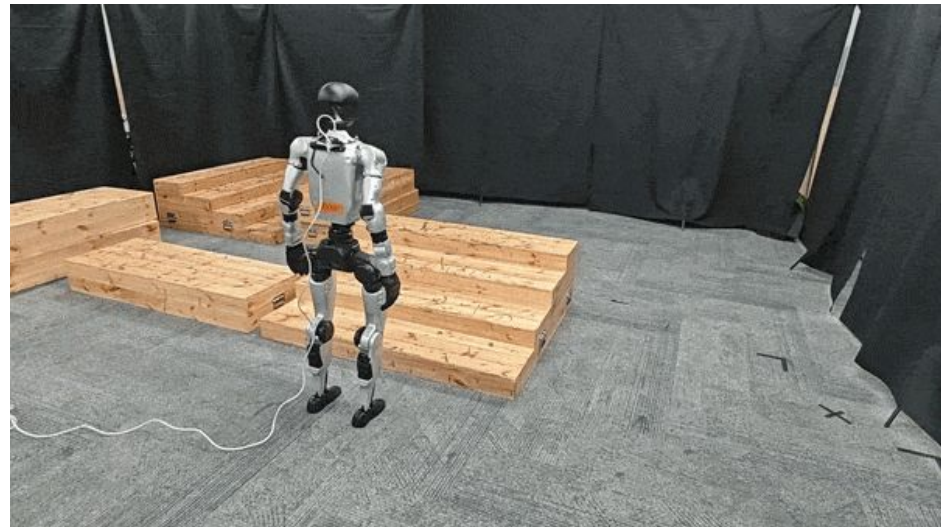
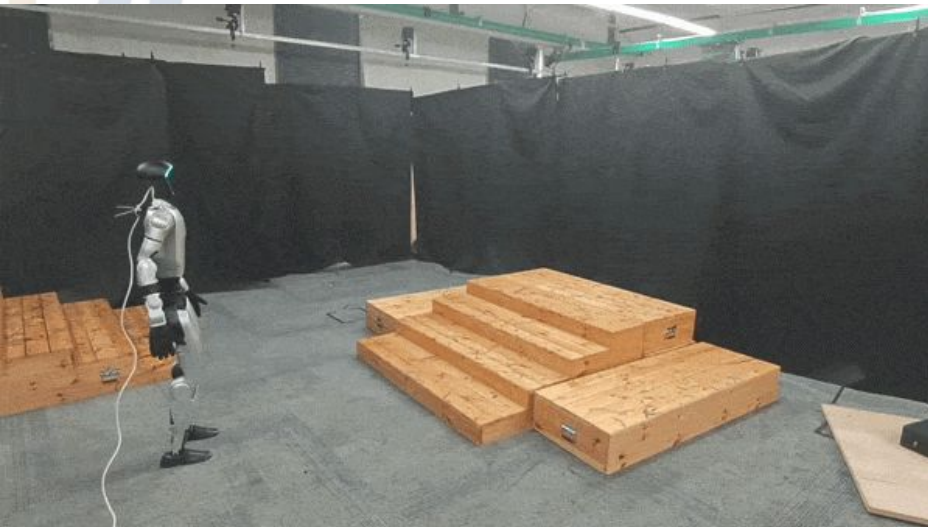
- Urmson, Chris, et al. "Tartan racing: A multi-modal approach to the darpa urban challenge." (2007).
- ROS tutorials: <https://www.ros.org/blog/getting-started/>



Thank you!

Perceptive Behavior Foundation Model

- We can even do backflips while walking up stairs!
 - Original joint commands are for flat grounds.



- More videos : <https://acodedog.github.io/perceptive-bfm/>